

Hansen's paper

$$3\mathbb{E} \quad \mathbb{E} \quad \mathbb{E}$$

§1. Transform (T)

Classical idea. $G_1 \times G_2 \curvearrowright V$

$$\rho \in \text{Rep}(G_1) \rightsquigarrow \text{Hom}_{G_1}(V, \rho) =: T(\rho) \in \text{Rep}(G_2)$$

$$\text{Tr} T(\rho)(g_2) \stackrel{?}{=} \int_{G_1} \text{Tr}(V)(g_1, g_2) \cdot \text{Tr} \rho(g_1) dg_1$$

Problem. ① V is from geometry

may not be compactly supported + $\text{Fix}(g_1, g_2)$ non-discrete

② Need to handle $\int_{G_1} = \infty$.

Solution. - Consider "compact" g_2 .

- Use test function $f \in C_c^\infty(G_2)$ (distribution)

F p -adic local field, G/F reductive group, $b \in B(G) = G(\check{F})/\sigma\text{-conj.}$

$$\Gamma = \text{Gal}(\check{F}|F)$$

$$\mu: G_m \rightarrow G/F^{\text{alg}} \longleftrightarrow \forall \mu \in \text{Rep}(G^\vee)$$

local shtuka space

$$\text{Sht}_{G, b, \mu} \rightarrow \text{Spd } C, \quad C = \mathbb{A}_p$$

$$S \mapsto \left\{ \beta: \mathbb{G}_m^{\times b} \rightarrow \mathbb{G}_m^{\times \mu} \text{ modification of } G\text{-bundles on } X_S \right\}$$

① β iso. on $X_S \setminus D_S$
 $\hookrightarrow$ (arbitrary div. for $S^\#$)

② β bounded by μ pointwise on S $= X_S^*(\Gamma)$

$$\hat{U}_{X_C^b, X_C} \simeq \widehat{B_{dR}^+(C)}, \quad B_{dR} = \text{Frac}(B_{dR}^+), \quad \beta_C \in G(B_{dR}^+) \backslash G(B_{dR}) / G(B_{dR}^+)$$

$$\beta_c \leq \mu, \text{ i.e. } \mu - \beta_c \in R_+$$

Then $G(F) \times G(F) \curvearrowright \text{Sh}_{G,b,\mu}$

$b=1$: $G(F) \times G(F) \curvearrowright \text{Sh}_{G,1,\mu}$
 $\in B(G, \mu)$

period map

$$\begin{array}{ccc} & \text{Sh}_{G,b,\mu} & \\ \pi_1 \swarrow & & \searrow \pi_2 \\ \omega_{G,\leq \mu}^b & & \omega_{G,\leq \mu}^1 \end{array}$$

Hecke stack $\text{Hk}_G \xrightarrow{(h_1, h_2)} \text{Bun}_G \times \text{Bun}_G$
 $:= \{ b: \xi_1 \rightarrow \xi_2 \}$

B_{dR}^+ - affine grassmannian

$$\omega_{G,\leq \mu} = \frac{L^+G/L}{G(B_{dR})} \quad \begin{array}{c} \uparrow \quad \uparrow \\ G(B_{dR}) \quad G(B_{dR}^+) \end{array}$$

Schubert cell $\omega_{G,\leq \mu} = \omega_{G,\leq \mu}^1 = \left\{ (t, \xi) : \begin{array}{c} \uparrow \\ \text{Bun}_G \end{array} \begin{array}{c} b: \xi_{\text{triv}} \\ \cup \\ G(F) \end{array} \middle| \begin{array}{c} X_S \setminus D_S \rightarrow X_S \setminus D_S \\ \text{bounded by } \mu \end{array} \right\}$

Assume $b \in B(G, \mu)^{\text{basic}}$ from now on

Admissible locus $:= \{ (t, \xi) : \xi_c \simeq \xi^b \}$
 pointwise

$$\omega_{G,\leq \mu}^{1, \text{adm}} \subset \omega_{G,\leq \mu}^1$$

Th. 2 $\begin{array}{ccc} \text{Sh}_{G,b,\mu} & \longrightarrow & \text{Spd } C \quad (\{\text{triv}, \xi^b\}) \\ \downarrow & \lrcorner & \downarrow \\ \text{Hk}_{G,\leq \mu}^{b,1} & \xrightarrow{(h_1, h_2)} & \text{Bun}_G \times \text{Bun}_G \end{array}$

$$\begin{array}{c} G(F) \times G_b(F) \\ \curvearrowright \\ \pi_1: \text{Sht}_{G,b,\mu} \rightarrow \text{Gr}_{G,\leq \mu} \end{array} \quad \begin{array}{c} \curvearrowright \\ G_b(F) \end{array} \quad \text{is a } G(F)\text{-torsor on } \text{Gr}_{G,\leq \mu}^{b,\text{adm}}$$

$$\left(\beta: \xi^1 \rightarrow \xi^b \mapsto \underline{\beta(\xi^1)} \right)$$

$$\pi_2: \text{Sht}_{G,b,\mu} \rightarrow \text{Gr}_{G,\leq \mu}^1 \quad \text{is a } G_b(F)\text{-torsor on } \text{Gr}_{G,\leq \mu}^{1,\text{adm}}$$

Wr. If $x \in \text{Sht}_{G,b,\mu}$ fixed by $(g, g') \in G(F) \times G_b(F)$, then

$$\pi_1(x) \in \left(\text{Gr}_{G,\leq \mu}^{b,\text{adm}} \right)^{g'}, \quad \pi_2(x) \in \left(\text{Gr}_{G,\leq \mu}^{1,\text{adm}} \right)^g$$

Rem. $g \in G(F) \sim G(\text{BdR}) / G(\text{BdR}^+)$ computation done via Witt vector for computation of cohom. of
 g elliptic: $\left(G(\text{BdR}) / G(\text{BdR}^+) \right)^g_{\leq \mu} = \left(\text{Gr}_{G,\leq \mu}^{1,\text{adm}} \right)^g$ semi-infinite orbits

|
 computable via $T = C_G(g)$ elliptic max'l torus

Kornel. S_μ Satake sheaf on Hk_G .

$$T_\mu: D_{\text{ét}}(\text{Bun}_G) \rightarrow D_{\text{ét}}(\text{Bun}_G)$$

$$F \mapsto h_2! \left(h_1^* F \otimes S_\mu \right)$$

Th 1. Consider $\Pi_\mu: D(G_b(F)) \rightarrow D(G(F))$

$$p \mapsto i_1^* T_{V_\mu^\vee} (i_b)_! p$$

$$\text{here } i_1: [* / G(F)] \rightarrow \text{Bun}_G$$

$$i_b: [* / G_b(F)] \rightarrow \text{Bun}_G$$

Then $\Pi_\mu(p) \simeq R\Gamma(G, b, \mu)(p) := \varinjlim_{K \subset G(F)} R\text{Hom}_{G_b(F)} \left(R\Gamma_c(\text{Sht}_{G,b,\mu,K}, S_\mu), p \right)$

Goal. $b=1 \in B(G, \mu)^{\text{basic}} \iff V_\mu|_{Z(\hat{A})^\Gamma} = \text{triv} \quad (\exists \text{ only many such } \mu)$

Then. $\Pi_\mu: D(\text{Rep}(G(F))) \rightarrow D(\text{Rep}(G(F)))$

What is Π_μ ?

§2. elliptic $\longleftrightarrow$ cuspidal

$G(F)_{\text{SRS}} = \{ g \in G(F) : (A(g) \text{ max'l torus}) \}$ Zariski open
($\Rightarrow$ conn'd)

Recall. $T \subset G$ max torus is elliptic if $T/Z(A)$ an isotropic / F (split rank = 0)

Th 2. - $\exists$ elliptic max'l torus / F

- $G(F)_{\text{ell}} = \{ g \in G(F) : (A(g) \text{ ell. max. torus}) \}$ analytically open in $G(F)_{\text{SRS}}$.

E.g. $G = \text{SL}_2$, $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G(F)_{\text{ell}}$ iff $g \text{ SRS} + \text{Tr}(g)^2 - 4 \notin (F^\times)^2$

$(F^\times / (F^\times)^2 \text{ finite index, } (F^\times)^2 \text{ open in } F^\times)$

- Fix $M_0 \subset G$ minimal F -Len, $W_0^G = N_G(M_0)(F) / M_0(F)$.

$\mathcal{L}$ finite set given by $\mathcal{L} = \{ F\text{-Len } M \subset G \text{ s.t. } M_0 \subset M \}$
 $\bigcup_{M_0} W_0$

Then we have Weyl integration formula [Arthur 92] local trace

$f \in C_c^\infty(G(F))$, then $\int_{G(F)} f(x) dx = \sum_{M \in \mathcal{L}} \frac{|W_0^M|}{|W_0^G|} \int_{\Gamma_{\text{ell}}(M(F))} |D(r)| \cdot \text{Orb}(r, f) dr$

here $\Gamma_{\text{ell}}(M(F)) = M(F)_{\text{ell}} / M(F)_{\text{conj.}}$ with a measure.

$$\text{Orb}(r, t) = \int_{A_M(F) \backslash G(F)} f(x^{-1}rx) dx$$

$\uparrow$
split centre of M

$r \in \Gamma_{\text{eu}}(M(F))$

$$D(r) = \det(1 - \text{Ad}(r) |_{\text{Lie } \mathfrak{h} / \text{Lie } \mathfrak{h}_M(r)}) \quad \text{Weyl determinant}$$

$$dr \text{ on } \Gamma_{\text{eu}}(M(F)) \text{ is given by } \int_{\Gamma_{\text{eu}}(M(F))} \phi(y) dy := \sum_{\{T\}} \frac{1}{|W_T|} \int_{T(F)} \phi(t) dt$$

$\uparrow$
runs over eu. forms in $M/\sim$

Def 1 $f \in C_c^\infty(\mathfrak{h}(F))$ cuspidal if $\text{Orb}(r, t) = 0, \quad \forall r \in \mathfrak{h}(F)_{\text{rs}} - \mathfrak{h}(F)_{\text{eu}}.$

Th 2* f cuspidal function $\Leftrightarrow$ representable invariant distribution

$$\text{i.e. } \theta(f) = \int_{\mathfrak{h}(F)} \frac{\theta(x) f(x) dx}{\in L_{\text{loc}}^1}$$

θ smooth on $\mathfrak{h}(F)_{\text{rs}}$

$$\text{Then } \theta(f) = \sum_{\substack{T \text{ eu. forms} \\ \text{in } \mathfrak{h}(F)}} \frac{1}{|W_T|} \int_{T(F)} \theta(t) |D(t)| \text{Orb}(t, f) dt$$

Ex. $\forall \pi \in \text{Irr Rep}(\mathfrak{h}(F), \mathbb{C}), \quad f \in C_c^\infty(\mathfrak{h}) \mapsto \text{tr}(\pi(f)) \in \mathbb{C}$

Th 3 is representable by a locally integrable function $\theta(g)$ on $\mathfrak{h}(F)$.

$$\textcircled{1} \text{tr } \pi(f) = \int_{\mathfrak{h}(F)} \theta(g) f(g) dg$$

$\textcircled{2} \theta(g)$ is smooth on $\mathfrak{h}_{\text{rs}}(F)$

$\textcircled{3} |D(t)|^{1/2} \theta(t)$ is locally bounded on $\mathfrak{h}_{\text{ss}}(F)$.

Upshot. If $\text{tr } \pi(g)$ is computable $\forall$ elliptic g ,

then $\text{tr } \pi(f)$ is computable for cuspidal f .

Defn $\pi \in \text{Ker}^{\text{adm}}(\text{Rep}(G(F)))$ is non-elliptic if $\text{tr } \pi(g) = 0$, $\forall g \in G(F)_{\text{ell}}$.

Th 4 [HKW, Th C.1.1] π nonelliptic $\Leftrightarrow \pi \in \text{span of proper parabolic inductions}$.

Proof $\Leftarrow$: van Dijk's formula

$\Rightarrow$: Kazhdan density thm [Kaz 86]

§3. gradings & inductions (via distributions)

Bernstein centre Distribution $D: C_c^\infty(G(F)) \rightarrow \mathbb{C}$

$$f \in C_c^\infty(G), \check{f} \in C_c^\infty(G): \check{f}(g) = f(g^{-1})$$

$$\check{D} \text{ by } \check{D}(f) := D(\check{f}).$$

$$(D * f)(g) = \check{D}(g \cdot f), \quad g \cdot f(x) = f(xg)$$

Def'n D essentially compact if $D * f \in C_c^\infty(G)$, $\forall f \in C_c^\infty(G)$.

D G -invariant if $D(gf) = D(f)$, $gf(x) = f(g^{-1}xg)$

Th 5 (Bernstein centre)

$$(P_1 * P_2)(f) = \check{D}_1(P_2 * f).$$

$$\mathcal{Z}(G) \simeq \{G\text{-inv. ess. compact distributions on } G(F)\} \xrightarrow{\text{faithful}} C_c^\infty(G)$$

Eg. $\phi \in C_c^\infty(G)$, $D_{\phi, dx}(f) := \int_G f(x) \phi(x) dx$

Spectral side $\pi \in \text{Rep}^{\text{adm}}(G(F))$, $D = (f \mapsto \text{tr } \pi(f))$

Geometric side $\gamma \in G_{\text{rs}}(F)$, $D = (t \mapsto \text{Ord}(\gamma t))$.

idea. spectral $\sim$ geometric.

$\swarrow$ K86, global trace formula

Th 6. (Kazhdan density thm) $f \in C_c^\infty(G)$, TFAE.

(i) $\text{tr}(\pi(f)) = 0$, $\forall \pi \in \text{Irr Rep}$

(ii) $\text{tr}(\pi(f)) = 0$, $\forall \pi$ tempered irrep

(iii) $\text{Ord}(\gamma, f) = 0$, $\forall \gamma \in G_{\text{rs}}(F)$

(iv) $f \in \mathbb{C}\text{-span}\{h^{-g}h\}$

Cor. $\{\text{Ord}(\gamma, -)\}_{\gamma}$ is dense in all G -inv. distributions

Rem. Trace Paley-Wiener Thm [BDK86] about image.

$$\left[\pi \mapsto \text{tr} \pi(f) \right] \quad \downarrow \quad \text{using parabolic induction.}$$

Cor. $\forall M \subset G$ Levi, $\exists$ canonical map $r_M: \mathcal{Z}(G) \rightarrow \mathcal{Z}(M)$

Def. $I(G) = C_c^\infty(G) / C_c^\infty(G)^{\text{null}}$

$\downarrow$
 $f \mapsto \text{tr} \pi(f) = 0$.

$\exists$ canonical map $r_M: I(G) \rightarrow I(M)$.

$f \mapsto f_M$.

s.t. $\text{tr}(\pi(f_M)) = \text{tr}(i_M^\pi \pi(f))$, $\forall \pi \in \text{Irr Rep}(M(F))$.

Prop ①
$$\begin{array}{ccccc} (\mathfrak{z} + f)_M & = & r_M(\mathfrak{z}) * & f_M \\ \uparrow & \uparrow & \uparrow & \uparrow \\ \text{dist} & C_c^\infty(G) & \text{dist} & C_c^\infty(M) \end{array}$$

② f cuspidal $\Leftrightarrow f_M = 0, \forall M \subsetneq G$

$$\text{Dist}(G) = I(G)^* = \{ G\text{-inv. distributions on } G \}$$

Then $\mathcal{Z}(G) \simeq \text{Dist}(G)$ faithful.

$$D(G) = \text{Span}(\pi(f))_{f \in \mathcal{Z}(G)} \subset \text{Dist}(G)$$

$$D^{\text{temp}}(G) \subset D(G)$$

$$\begin{array}{ccc} D^{\text{temp}}(G) & \hookrightarrow & \mathcal{Z}(G) \\ \downarrow & & \downarrow \\ 0 & & \mathcal{Z} \end{array} \quad \begin{array}{c} \text{faithful} \\ \uparrow \\ K \text{ density th.} \end{array}$$

$$(\mathcal{Z} * \theta)(f) = \theta(\mathcal{Z} * f)$$

Th 7 (Arthur 93, 96)

$$\begin{array}{l} I(G) = \bigoplus_{M \in L/W_0} (I(M)^{\text{usp}})^{W(M)} \\ \mathcal{Z}(G) \hookrightarrow \\ \subset D^{\text{temp}}(G) = \bigoplus_{M \in L/W_0} \underbrace{(D^{\text{ell}}(M))^{W(M)}}_{\substack{\uparrow \\ \text{elliptic temp.}}} \end{array}$$

Let $(H) \in D^{\text{temp}}(G)$, then $(H) = (H)^{\text{usp}} + \underbrace{(H)^{\text{ind}}}_{\substack{\uparrow \\ \text{parabolic induction}}}$

$$f \in I(G)$$

$$b = f^{\text{usp}} + f^{\text{nc}}$$

Obs. of [H]. $\mathcal{Z} \in \mathcal{Z}(G)$, $\Pi_\mu: D^{\text{temp}}(G) \rightarrow D^{\text{temp}}(G)$

Th 8. If ① $\Gamma_M(\mathcal{Z}) = \text{"very stable"}$ $\forall M \subsetneq G$. ② $\mathcal{Z}$ commutes with $\{\Pi_\mu\}$, $\forall \mu$
 $\xRightarrow{\text{average}} \mathcal{Z} \text{ "very stable"}$